Automated Detection and Classification of Banana Leaf Diseases Using Convolutional Neural Network With Spatial Attention Mechanism

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***Abstract*—Banana is a traditional crop that helps in ensuring food security of the world in general and local economy in particular. Its production is however threatened by several leaf diseases that lead to major losses of production. Traditional techniques of detecting diseases are normally cumbersome and slow and also requires a professional knowledge and therefore it cannot come in time to intervene. This paper proposes an automated, deep learning model which is used to precisely detect and classify four primary conditions of the banana leaves as: Sigatoka, Cordana Leaf Spot, Pestalotiopsis Leaf Blight, and Healthy. To highlight the pathologically active features, Convolutional Neural Network (CNN) model with a spatial attention mechanism named as KernelAttention is developed. A well-selected dataset was used to train and test the model with 937 original and 1600 augmented images. Manual curation of the dataset with 1,351 training, 120 validation and 760 test images is done to train the model. The model achieved a high overall test accuracy of 97.50%. Particular performance metrics which are precision, recall, F1-score, and specificity on all classes were better than 0.93 and most classes have AUC-ROC of above 0.999. That would be indicative of exceptionally high discriminatory power. Moreover, the use of Gradient-weighted Class Activation Mapping (Grad-CAM) illustrations proves that the model is open for interpretation as they depict that the model generates biologically meaningful predictions regarding disease symptoms. The high predictive accuracy and the visual interpretability of proposed CNN with spatial attention mechanism makes it a stable and legitimate tool of early and efficient diagnosis of banana leaf diseases and hence presents the gateway to a better picture of crop management and reduction of crop losses.**

***Index Terms*—Deep Learning, Convolutional Neural Network (CNN), Spatial Attention Mechanism, Image Classification, Banana Leaf Disease**

I. Introduction



Fig. 1. Healthy Banana Leaves

As shown in Fig. 1, Banana (Musa spp.) is one of the most economically relevant fruit crops that can be found all over the world producing an important source of food and a major source of income to millions of people especially in the tropical and subtropical destinations. [1] It has great nutritional value and is available year-round, which earns it major contribution to food security in the world. [2] However, banana production is under continuous threat of various plant pathogens, most of which are fungi, that may cause disastrous losses of yield and economic ravage to the farming communities. [3] Detecting such diseases at their early stages in accurate and efficient ways is thus, a dire requirement, in the application of effective and timely management conditions, and subsequently reduce degradation of crops and enhance sustainable agricultural processes. Among the various diseases affecting banana foliage, Sigatoka, Cordana Leaf Spot, and Pestalotiopsis Leaf Blight represent particularly virulent threats. [1] [4]



Fig. 2. Sample banana leaves affected by Sigatoka Disease

Sigatoka Fig. 2, primarily caused by Mycosphaerella fijiensis, reproduces its effects in the initial stages as small and dark brown or black spots on the leaf surface. The lesions enlarge, coalesce and the leaves then turn yellow and dry up and die. This disease is highly destructive and can reduce fruit yields by 50% or more if left uncontrolled. It also spreads through air, wind, rain and through use of farm tools which are contaminated as well as contact with human beings. It thrives easily in hot and humid climate areas that receive abundant rainfall. [1] [4]



Fig. 3. Sample banana leaves affected by Cordana Disease

Cordana Fig. 3, caused by Cordana musae, forms circular, water-soaked spots that later become brown and desiccated. It can lead to severe symptoms which result in premature defoliation, causing a significant loss of photosynthetic ability and directly affecting fruit production. It spreads through infected planting materials, water, and air, especially in humid climates. [1] [4]

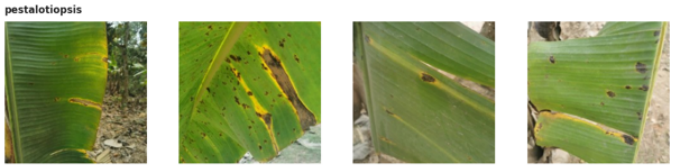


Fig. 4. Sample banana leaves affected by Pestalotiopsis Disease

Pestalotiopsis Fig. 4, caused by Pestalotiopsis spp., is mainly attributed to the existence of irregular brown patches which in most instances develop their occurrences at water-soaked patches. The lesions which form become inflamed and merge together to form necrosis on the surface of whole patches of the leaf. Transmission of this pathogen is commonly carried out by way of the equipment utilized in the agricultural sector in addition to the leaf refuse material and also the vegetable goods. It finds its most favourable conditions in damp, warm, and ill-aerated environments. It is a serious disease which is responsible for weakening of plant health. [1] [4]

Differentiating of these different diseases, and their clear distinction on the healthy foliage, using only visual symptoms, is one of the main prerequisites to implement effective and targeted disease management. Traditionally, diagnosis of banana leaf diseases has relied on manual visual inspection by experienced plant pathologists or other advanced lab-based molecular and microscopic tests. Although these traditional techniques are accurate, are labour-intensive and expensive for large scale farming or in areas where resources are limited or is not feasible due to the factors, such as their dependence on specialized skills, the cost involved in operation and impracticality to use in large scale farming operations. This delay in such diagnostic methods can result in a rapid advancement of the disease as well as a massive contamination, and an irreversibly damaged crop, which therefore necessitates an immediate need to find effective, scalable, and automated detection methods. [5]

The new epochal breakthrough in computer vision and deep learning that has occurred has radically transformed most sectors especially in the agricultural sector by availing of innovative opportunities in the automatic detection of plant diseases. [6] In specific, Convolutional Neural Networks (CNNs) have been discovered to emerge as the powerful image based classification model networks as far as the use of images is concerned which is helpful in classification. [7] It is their very nature of being able to learn representations of more detailed and more abstract properties directly and completely out of the crude raw pixel data byproducts of lengthy and even subjective hand generated feature engineering. Implementation of systems which have the potential to diagnose the plant diseases fast and accurately through digital pictures of leaves can be carried out by adopting the unmatched powers of CNNs. [8] [9] Such systems will reduce the dependency on the scarce human expertise quite significantly and will allow timely interventions to safeguard productivity in the farming sector.

This paper implements a deep learning framework that can be able to classify the significant banana leaf diseases precisely. Primary contribution in this paper is the creation of architecture of a Convolutional Neural Network integrated with Spatial Attention Mechanism [10] [11] named as KernelAttention. This custom design is well executed so as to enhance the discriminative power of the model such that it focuses more on the most salient symptomatic parts of the captured image of the leaf selectively, or adaptively. The model is evaluated using a carefully collected dataset of banana leaf images obtained under natural farming conditions. The precision is attentively estimated with the help of a complete range of standard assessment metrics, including precision, recall, F1-score, specificity, and detailed ROC-AUC. Additionally, we obtain Grad-CAM (Gradient-weighted Class Activation Mapping) visualizations that help to get the important information about the reasons, which made the model choose such way of decision. Having determined the certain capabilities of the model to consider the biologically meaningful disease markers, it is also possible to say that the model is thus

highly applicable. Findings reveal how smart visual systems can motivate new ways of working in disease management in banana agriculture, and such breakthroughs will automatically cross over to new horizons in food security and sustainable agriculture.

II. RELATED WORKS

BananaSqueezeNet proposed by Bhuiyan et al. [1], is a very fast and lightweight CNN model optimized through Bayesian Optimization which achieved benchmark accuracy of 96.25 percent, and precision 96.53 percent, recall 96.25 percent, specificity 98.75 percent, F1-score 96.17 percent, and MCC 95.13 percent. The author Rajalakshmi et al. (2025) [2] classified banana leaf diseases by the eight-layer DCNN. Model was trained on 768 original images, with the accuracy of up to 98.92%. Precision and recall and F1-score being above 99 percent. The model consisted of both convolution and pooling layers, Batch normalization and global average pooling layers was used to enhance its learning process to avoid overfitting.

Ünal [3] incorporated deep learning into diagnosis of banana leaf spot diseases where DenseNet- 201 achieved an accuracy of 98.12 percent, VGG16 achieved accuracy of 97.81 percent and EfficientNet-b0 achieve 87.81 percent accuracy. Helmawati and Utami, [8] [12] trained CNN on Cordana, Pestalotiopsis, Sigatoka, and Healthy leaves and attained the accuracy of 92.85 percent and an accuracy of 90.62 percent respectively. Syihad et al. [13] applied CNN based Approach Using ResNet50 and VGG19 as methods to detect the disease on banana plants using banana leaves images. Their experiments showed ResNet50 reached a good metric of 94 percent accuracy, and VGG-19 achieved 94 percent accuracy.

Elinisa et al. [14] evaluated U-Net image segmentation architecture using deep learning neural network in semantic segmentation. The average score obtained on each of the U-Net groups was 96.45 percent Dice coefficient and 93.23 percent Intersection over Union in segmentation of both the diseases Fusarium Wilt and Black Sigatoka on the banana plants.

Kim et al. [15] introduced Spatially Attentive Output Layer (SAOL), a new architecture however, the output layer that involves the use of spatial attention is incorporated in its place compared to the explicit global average pooling (GAP) followed by a fully connected block (FC). The authors demonstrate that in various CNNs on CIFAR-10, CIFAR-100 and ImageNet, SAOL outperforms traditional output layers by up to +1.77 percent top-1 accuracy and with only a slight overhead in FLOPs (5-6 percent across the experiments), at constant training load. In addition, the maps of attention which are represented by the model SAOL introduces competitive weakly- supervised object localization (WSOL) which does not need any backward pass or subsequent localization layers.

Sanga et al [16] used ResNet152 and InceptionV3 models to detect Fusarium Wilt Race 1 and Black Sigatoka on 3,000 banana leaf images in early detection of any disease as well. The Accuracy results of ResNet152 was 99.2 percent and that of InceptionV3 was 95.41 percent. InceptionV3 was chosen for mobile deployment because of its reduced memory footprint even though it has slightly lower accuracy. The application detected both diseases with confidence level of 99 percent through the captured leaf area, which could be applied to enhance yields among small holder banana farmers by availing disease interventions.

Arifin [17] CNN GoogleNet architecture algorithm is considered to classify disease in banana leaves and segregate it into a particular category. This analysis was categorized into 3 groups, the healthy leaves, sigatoka and Fusarium wilt. As a brief description, GoogleNet consists of five to thirteen layers. Based on the findings of the analysis of the trial process which has been done at the research stage from epoch 10 to 100 it could be concluded that GoogleNet yielded the best accuracy results model at batch size 35 with epoch 60 for Healthy leaf class 90.59 percent, with epoch 80 for Fusarium Wilt 90.30 percent and at 90.49 percent in sigatoka.

In Automated Segmentation and Classification model With CNN, the authors Krishnan et al. [18] introduce a two-step system: to begin with, Total Generalized Variation Fuzzy C-Means (TGVFCMS) should segment the diseased areas of the banana leaf with the aim of refining the edge sharpness and removing the noise. These segmented regions then undergo a convolutional neural network (CNN) that can predict the disease, based on learnt features such as color, texture, and shape. The TGVFCMS + CNN proposed model achieved classification accuracy of 93.45 percent. Ashurov et al. [19] proposed Depthwise CNN with Squeeze and Excitation blocks and addition of Residual Skip Connections which has shown very high accuracy of 96 percent and F1 scores of 98 percent, making it one of the first papers to automate disease identifications in plants.

Kumar et al. [20] proposed Spatial Attention-guided pre-trained networks for accurate identification of crop diseases. Authors present a modified EfficientNet-B3 backbone (EfficientNetB3-ACSA) with two important modules, including an ancillary convolutional layer that generates fine-grained local learned features and a spatial attention component that learns what areas of the image are important (such as spots or lesions). This model is trained and tested on a combination of 3 datasets (PlantVillage, Tomato-Village, a hybrid field-and-laboratory data with library and diseased tomatoes over 38 disease classes, and more than 100,000 labeled images in total). The suggested model performs impressively, with 99.89 percent accuracy, with near perfect precision, recall and F1 scores, beating other state-of-the-art CNNs and hybrid Transformer based models [21].

ANN+SIFT Model has been proposed by Thiagarajan et al [22]. Based on the study, the authors have proposed and developed a machine learning formulation that aims to address the challenge of identifying the banana leaves disease using images which are at different angles and distances. The researchers tested this ANN-SIFT model on BananaLSD dataset comprising of 937 images of healthy leaves and infected leaves of three diseases, Pestalotiopsis, Sigatoka and Cordana. Model

achieved an accuracy of 95 percent in their determination of the health status of the plant.

The research conducted by Alanazi et al. [23] used YOLOv10 model in determining the diseases and classification of many conditions leading to the destruction of banana leaves. YOLOv10 was also a concise algorithm in detecting and identifying such conditions on the banana leaves. It also depicts an efficient picture classification and localization of the conditions with a mean average precision (mAP@0.5) of 88.85 percent. In addition, the model accuracy rate is 91.22 percent and high recall of 85.06 percent.

The study by Guo et al. [24] shares a methodology where the framework is separated into various stages the aim of which is to identify plant diseases through the images taken in a real environment of an agricultural field. Architecture of the model consists of the three step pipeline. Its objective is to identify a leaf in a cluttered background and cull it by making use of a Region Proposal Network (RPN) then Chan-Vese (CV) algorithm and finally, the pre-trained ResNet-101 deep learning model. The principal result that has been released this entire three-stage pipeline achieved test accuracy of 83.57 percent on the images within a complex environment. Performance of the model also varied on different types of diseases, with 90 percent accuracy on rust, 80 percent on the plaque bacteria and 75 percent on black rot.

Aruraj et al. [25] introduced an image processing approach based on LBP to efficiently determine and classify some of the diseases such as Black Sigatoka and Cordana leaf spot in banana plants which can increase agricultural productivity in Asia and Africa. In this study, techniques in image processing with special attention to LBP technique were used along with machine learning algorithms such as SVM and KNN to classify diseases in banana plants. The findings indicate an 89.10 percent accuracy rate for Black Sigatoka and a 90.90 percent accuracy rate for Cordana leaf spot classification.

III. Methodology

The methodology used to in this model is provided in this section including all the procedures involved in the collection and characterization of the dataset, architectural complexity and design insights of proposed Convolutional Neural Network (CNN) with spatial attention mechanism and detailed processes of training process of the model, method of optimization and accuracy.

A. Dataset Overview

The quality and variability of a dataset are important aspects that drive any deep learning model's performance. Therefore, this research used a specifically good image dataset of banana leaves with variability in real agricultural settings. For this study, we used the BananaLSD dataset, which contains images taken in uncontrolled, natural field conditions.

Unlike laboratory-curated datasets such as PlantVillage [26], where images are collected in highly controlled environments with uniform backgrounds and consistent lighting, BananaLSD [27] incorporates the natural variability of real-world farming contexts. These include differences in lighting intensity, shadows, background clutter, and environmental noise. Training a model on such realistic data ensures better robustness and generalization, as the network learns to handle the same variability it will encounter in actual deployment. As a result, BananaLSD is a more relevant and reliable dataset for building a model designed to be utilized by farmers and agricultural professionals in the field.

All images in the dataset were labelled by trained plant pathologists who provided ground truth annotations that were very high quality. Having high quality ground truth annotations of our images will add to the reliability of our diagnostic system and build confidence in the predictions of the trained model. The data set also adopted a fixed split for training, validation and testing; the test set was purely based on original images.

B. Dataset Composition and Characteristics

The data used in the study is a contribution by the BananaLSD project [27], carefully gathered from two primary geographic locations: the dedicated banana farm associated with Bangabandhu Sheikh Mujibur Rahman Agricultural University, and nearby farmlands of local farmers in the Gazipur, Bangladesh area. Image acquisition was conducted in June 2021. The dataset is completely labeled and contains images representing four different classes of conditions that can occur on banana leaves: Pestalotiopsis Leaf Blight, Sigatoka, Cordana Leaf Spot and Healthy leaves. The first set consisted of 937 original RGB JPG images. The set of images were collected by hand with the use of a set of three different smartphone cameras. To facilitate compatibility of the images with CNN architectures, all images were given the same common image size of 224x224 pixels where all images had been resized to the same form. The dataset is publicly available on Mendeley Data [4].

From Table II, the distribution of dataset across the four defined classes was observed to be imbalanced where it is seen that Sigatoka has largest representation and Healthy leaves have lowest percentage.

TABLE I
ORIGINAL DATASET COMPOSITION BY CLASS

Name of Disease	Original Images	Augmented Images
Pestalotiopsis Leaf Blight	173	400
Sigatoka	473	400
Cordana Leaf Spot	162	400
Healthy	129	400
Total	937	1600

C. Data Augmentation and Preprocessing

The BananaLSD dataset originally showed a marked class imbalance as indicated in Table I. Such an imbalance can lead the model to be biased toward the majority classes, causing the model to poorly generalize and under-represent the minority classes in evaluation times.

To solve this imabalance, data augmentation was used. Augmentation helps not only to balance the dataset, but to diversify it as well. Data augmenting allows to limit overfitting during training and improve the models generalization beyond training to images never seen before.

A combination of geometric and photometric transformations were used with a focus on improving the robustness of the models. The augmentation process was considered specifically through each transformation. Random rotation and flipping horizontally ensured that the model would ultimately be invariant to the orientation of leaves in the field. Width and height (translation) shifting introduced horizontal and vertical shifts to represent differences in leaf positioning during image capture. The shear transformation on the dataset attempted to mimic natural images distortion due to camera angles and leaf bending. Zooming allowed the model to be invariant to the distance of the images being captured (i.e., close or far) of the leaf. Lastly, a fill mode strategy was used for the empty pixels created from the transformations to keep visual integrity throughout the transformations.

After augmentation, the dataset was divided into training, validation, and test subsets. The core principle of this method is that both the validation and test subsets contain only original images to maintain an unbiased evaluation. Augmentation was only applied to the training data for classes to maintain an admissible balance in the class distributions. As shown in Fig. 5, the final design was targeted to have 350 training images and 30 validation images and 190 test images per class. Although the Sigatoka and Pestalotiopsis classes had the targeted number of training samples, the Cordana and Healthy classes had slightly less for augmentation feasibility purposes (342 and 309 samples respectively). However, this approach ensured a justifiable balance of the evaluation subsets with enough and proportional samples for each disease class.

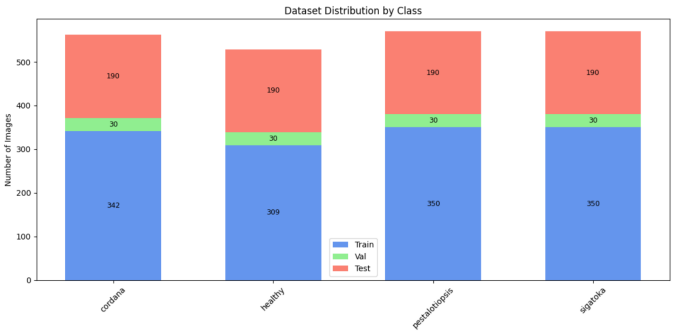


Fig. 5. Dataset Distribution by Class

D. Model Description

The deep learning model proposed to achieve high-specificity classification of banana leaf diseases is a Convolutional Neural Network (CNN) integrated with a spatial attention mechanism. An important aspect of proposed design is a dedicated spatial attention mechanism [10] named KernelAttention. The spatial attention mechanism allows the network to dynamically and adaptively direct attention to spatial regions dense with information within the extracted feature maps. This attention measures the extent to which the model has encoded attention at the highest level to the most diagnostically important areas of the input, amplifying the presence of symptoms of disease and the different diagnostic characteristics those symptoms entail, in order to maximize accuracy of classification. The model receives input images of size $224 \times 224 \times 3$, which correspond to 224 pixel width and height (224 x 224) RGB images of banana leaves and 3 associated channels. The full model architecture comprises an initial convolutional block, three KernelAttention blocks that follow consecutively, and a classification head. Each sub-structure increasingly diminishes low-level details while expanding the receptive field through multi-scale, multi-level feature maps primary visual experiences simultaneously. This construction allows the network to learn high-level patterns of visual features that can accurately be mapped both to identifying and differentially diagnosing banana leaf diseases.

KernelAttention is intentionally equipped with three core attributes in mind. First, a zoned spatial emphasis is created by dynamically modifying spatial attention gradients such that the areas of the leaf most indicative of disease are given greater attention (e.g., as compared to channel-based mechanisms such as Squeeze-and-Excitation (SE) [19] [28], which are primarily designed to effectively capture inter-channel relationships). The use of spatial emphasis in attention may be most effectively used in the localization of disease patterns that can be identified in specific area of the leaf. The second feature is that KernelAttention can apply attention in hierarchical distributions across the entire network instead of only on the output of the network (e.g., the Spatially Attentive Output Layer (SAOL) [15] approach). The hierarchy allows importance for spatial attention to be adjusted at deeper abstraction levels of feature importance, allowing the model to reflect advanced spatial relationships at varying scales. The last feature is that it is provided with explicit mathematical notation that explains how the features for query, keys, and values are created, how the attention map is fused additively and via element-wise multiplication, and how residuals are applied. The different approaches provide a clearer understanding of how attention is created and fusing concepts and allows others to understand our proposed method and provide flexibility for others to replicate and build off our efforts.

The structure of the network architecture is described as follows(Fig. 6):

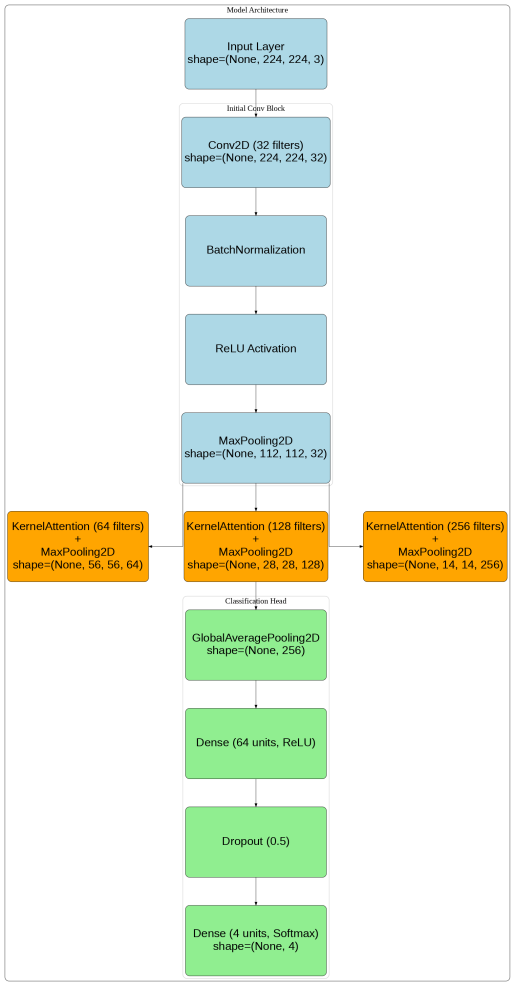


Fig. 6. Block Diagram of Model Architecture

As shown in Figure 6, the block diagram illustrates the overall architecture of the proposed model.

1) Input Layer: The model receives its raw input as an RGB image with dimensions of 224×224 pixels. This forms the initial feature map $X_0 \in \mathbb{R}^{224\times224\times3}$.

2) Initial Feature Extraction Blocks: Convolutional Layer: First is a 2D Convolutional Layer, consisting of 32 filters each with dimensions 3×3, The 'same' padding will preserve the spatial output size and (assuming stride=1) allows this layer to maintain the same input resolution whilst detecting basic visual features such as edges, textures and basic shapes in the images of the leafs.

Batch Normalization: Following the convolution, a Batch-Normalization layer will subsequently be applied. This approach is used to stabilising and speed up training by normalising the activations of the preceding layer for each mini-batch to address covariate shift, and the shared weights of each processing of each mini-batch.

Activitation Function: Next, a ReLU (Rectified Linear Unit) will be applied in an element-wise operation. This non-linear activation function provides the model with sufficient non-linearity, which is essential for the model to learn complex non-linear relations that exist in the data.

Max Pooling: A Max-Pooling layer with a 2×2 pool size is used. This process reduces the size of the feature maps by taking the highest value from each 2×2 window. This downsampling lowers the computational load and makes the model less dependent on the exact location of features in the image.

3) Stacked KernelAttention Blocks (Spatial Attention Integration):

The primary advancement of this model, as well as its ability to utilize spatial awareness, is created by stacking three KernelAttention blocks. Every block develops and applies a spatial attention map to determine the relevance of diverse schemes to the feature maps. This metric essentially redistributes the model's representation capability to the most diagnostically relevant visual evidence. Then a Max Pooling Layer continues to down-sample the feature maps and the Hierarchical Representation is refined sequentially with each KernelAttention block.

First KernelAttention Block: The output of the first extraction block as feature map is the input to the first KernelAttention layer with 64 filters. We illustrate how this specialized layer operates internally in Figure 7. It abstracts a feature map from spatial attention maps which it then performs element wise multiplication. The resulting attention boosted output is then down sampled by a Max-Pooling Layer of size 2×2.

Second KernelAttention Block: The refined features processed by the first layer, feed-forward to the succeeding second layer with 128 filters, which is itself a KernelAttention layer. At this stage progressively more detail continues to be abstracted from the input image with stronger and stronger spatial attention. Next Max-Pooling Layer is again of 2×2 2×2 size is followed.

Third KernelAttention Block: The third block has 256 filters which again operate at the highest level of abstraction and only selectively enhance important information necessary for precise disease classification of the scanned images. The last pooling layer is the last A Max-Pooling Layer with a size of 2×2 to complete spatial dimensionality reduction. It has a crucial function before feature representation moves on to classification head process.

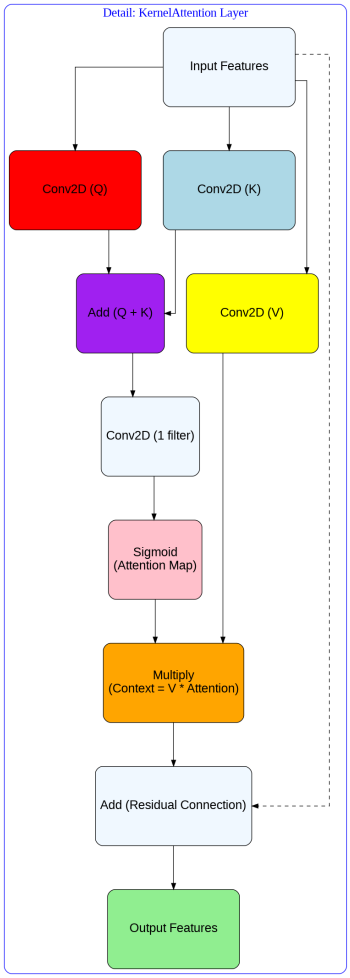


Fig. 7. Block Diagram Of Kernel Attention Layer

IV. Mathematical Formulation of the Model

This section provides a detailed mathematical characterization of the proposed architecture of the Convolutional Neural Network (CNN), describing the basic operations of the different layers, and parameterizing the operation of the KernelAttention spatial attention layer that is customized.

A. Standard CNN Layer Operations

The underlying components of this model consist of standard deep learning layers, each of which applies a specific transformation on the input for the purpose of feature extraction.

1) Convolutional Layer (A 2D convolutional layer): For an input feature map $X \in \mathbb{R}^{H\times W\times C_{in}}$, and a filter $F_k \in \mathbb{R}^{h_k\times w_k\times C_{in}}$, the output feature map $Y \in \mathbb{R}^{H'\times W'\times C_{out}}$ at position (i, j) is:

$$Y_{ijk} = \sum_{a=0}^{h_k-1}\sum_{b=0}^{w_k-1}\sum_{c=0}^{C_{in}-1} X_{i\cdot s_h+a,\, j\cdot s_w+b,\, c}\cdot (F_k)_{abc} + B_k \tag{1}$$

where s_h, s_w are strides, B_k is the bias term, and padding='same' ensures $H' \approx H$, $W' \approx W$.

The filter F_k is a learnable pattern detector, which is shifted across the input to generate a feature map. The feature maps will contain visual patterns including edges, textures, or particular crop leaf disease lesions to allow the network to learn diagnostic features automatically.

2) Batch Normalization (BN): Given a mini-batch $\mathcal{B} = \{x_1, \ldots, x_m\}$:

$$\mu_{\mathcal{B}} = \frac{1}{m}\sum_{i=1}^{m} x_i \tag{2}$$

$$\sigma^2_{\mathcal{B}} = \frac{1}{m}\sum_{i=1}^{m}(x_i - \mu_{\mathcal{B}})^2 \tag{3}$$

$$\hat{x}_i = \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma^2_{\mathcal{B}} + \epsilon}} \tag{4}$$

$$y_i = \gamma\hat{x}_i + \beta \tag{5}$$

where γ, β are learnable parameters and ϵ is a small constant for stability.

The batch normalization layer normalizes the output from the previous layer to maintain stability in the distribution while training. This results in faster convergence and improved learning stability, and reduces the implications of the initialization process. The parameters and aThe filter Fk is a learnable pattern detector, which is shifted across the input tγ and β generate a feature map. The feature maps will contain visual patterns including edges, textures, or particular crop leaf disease lesions to allow the network to learn diagnostic features automatically. llow the network to intelligently restore a desired optimal distribution, when required.

3) Rectified Linear Unit (ReLU): The non-linearity is defined as:

$$\text{ReLU}(x) = \max(0, x) \tag{6}$$

The ReLU applies non-linearity by retaining positive values and replacing negative values with zero. This activation provides a simple and efficient form of non-linearity to counter the vanishing gradients in the earlier layers, while still permitting networks to learn complex, non-linear relations in the data.

4) Max Pooling: For pool size $p \times p$:

$$Y_{ijk} = \max_{a=0..p-1}\ \max_{b=0..p-1} X_{i\cdot p+a,\, j\cdot p+b,\, k} \tag{7}$$

with $p = 2$ in our model.

Max pooling lowers the spatial dimensionality of feature maps by selecting the highest activation in each local neighborhood. In general, max pooling has three positive benefits: it affords the model translation invariance, decreases computational demand, and mitigates overfitting by discarding information with limited importance.

B. Spatial Attention Mechanism

Given $X_{in} \in \mathbb{R}^{H\times W\times C_{in}}$, the KernelAttention layer applies the following operations.

1) Input Projection:

$$X_{proj} = \begin{cases} \text{Conv}_{1\times1}(X_{in}), & C_{in} \neq C_{out} \\ X_{in}, & C_{in} = C_{out} \end{cases} \tag{8}$$

The input projection ensures that the channel dimensions of the input match those of the output. If necessary, a 1×1 convolution is applied, otherwise the input is passed directly.

2) Query, Key, and Value Generation:

$$Q = \text{ReLU}(\text{BN}(\text{Conv}^{Q}_{3\times3}(X_{in}))) \tag{9}$$

$$K = \text{ReLU}(\text{BN}(\text{Conv}^{K}_{3\times3}(X_{in}))) \tag{10}$$

$$V = \text{ReLU}(\text{BN}(\text{Conv}^{V}_{3\times3}(X_{in}))) \tag{11}$$

The query (Q), key (K), and value (V) Projections from convolutional transformations are also normalized and passed through a non-linearity. The projections encode different views of the input and assist the attention mechanism by developing a basis for more or less spatial emphasis on which features to focus.

3) Spatial Attention Map:

$$A_{int} = Q + K \tag{12}$$

$$A = \sigma(\text{Conv}_{1\times1}(A_{int})) \tag{13}$$

where $\sigma(x) = \dfrac{1}{1+e^{-x}}$ is the sigmoid function.

The attention map A resulting highlight spatial areas that are diagnostically important. Because of the sigmoid activation to scale the maps between 0 and 1, the spatial maps provide normalization so that different areas of the input can retain more or less weight based on the importance to the classification problem.

4) Applying Attention and Residual Connection:

$$C = V \odot A \tag{14}$$

$$Y_{out} = C + X_{proj} \tag{15}$$

where $\odot$ denotes element-wise multiplication.

The attention-weighted features C projected spatially attended maps are summed with the project inputs by a residual connection, maintaining the original information and reinforcing that original information with a relative focus on the spatial aspect of the leaf area, resulting in a better learning ability of subtle patterns of disease.

5) Summary Expression: Finally, the operation can be summarized compactly as:

$$\begin{aligned} Y_{out} = {} & \big(\text{ReLU}(\text{BN}(\text{Conv}^{V}_{3\times3}(X_{in})))\big) \odot \\ & \sigma\Big(\text{Conv}_{1\times1}\big(\text{ReLU}(\text{BN}(\text{Conv}^{Q}_{3\times3}(X_{in}))) \\ & + \text{ReLU}(\text{BN}(\text{Conv}^{K}_{3\times3}(X_{in})))\big)\Big) \\ & + \text{Conv}^{opt}_{1\times1}(X_{in}) \end{aligned} \tag{16}$$

This final formulation concisely summarizes the overall KernelAttention process, by illustrating the manner in which it is derived from full embodiments, and how combining the value features with the spatial weights will continue to retain the information encoded into the original input through the residual projection. This processing approach allows the overall KernelAttention framework to utilize the projection maps, attention maps attached to the spatial emphasis (areas of leaves which are important to the disease), and the original input information into a final representation.

$$\begin{aligned} Y_{out} = {} & \big(\text{ReLU}(\text{BN}(\text{Conv}^{V}_{3\times3}(X_{in})))\big) \odot \\ & \sigma\Big(\text{Conv}_{1\times1}\big(\text{ReLU}(\text{BN}(\text{Conv}^{Q}_{3\times3}(X_{in}))) \\ & + \text{ReLU}(\text{BN}(\text{Conv}^{K}_{3\times3}(X_{in})))\big)\Big) \\ & + \text{Conv}^{opt}_{1\times1}(X_{in}) \end{aligned} \tag{17}$$

V. Results

All results of proposed model are displayed in this section. Evaluation of test set was done using different metrics such as accuracy, precision, recall (sensitivity), F1-score and specificity. In addition, the ROC curves and the AUC scores were calculated to get an ultimate value of the process of conducting classification, specifically, with respect to discriminating power. Finally, to explain model's visual explaination power, the Gradient-weighted Class Activation Mapping (Grad-CAM) was utilized.

A. Training Dynamics and Convergence

Figures 8 and 9 shows the training, validation, and test accuracy and loss metrics of the model, monitoring learning dynamics for convergence properties over a 40 epoch run. The plotted curves yield very useful information on the optimization process, model stability and the model's generalizing ability.

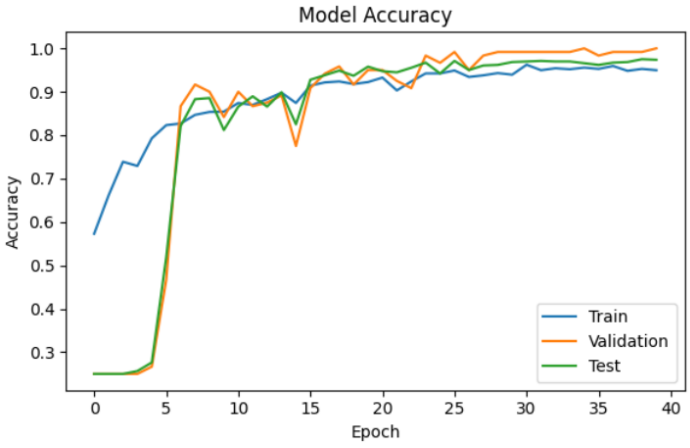


Fig. 8. Model Accuracy Graph

Model Accuracy Graph: Fig. 8 shows during the early stages of training all 3 curves (training - blue, validation - orange and test - green) were steeply increasing showing fast learning. Training accuracy reached nearly 100% in the last epochs, indicating that the model has well learned the train

data. More significantly, the patterns of validation accuracy and test accuracy were similar and were 99.17 percent and 97.50 percent, respectively, at 39th epoch. These results reflect strong learning while showing little over-fitting.

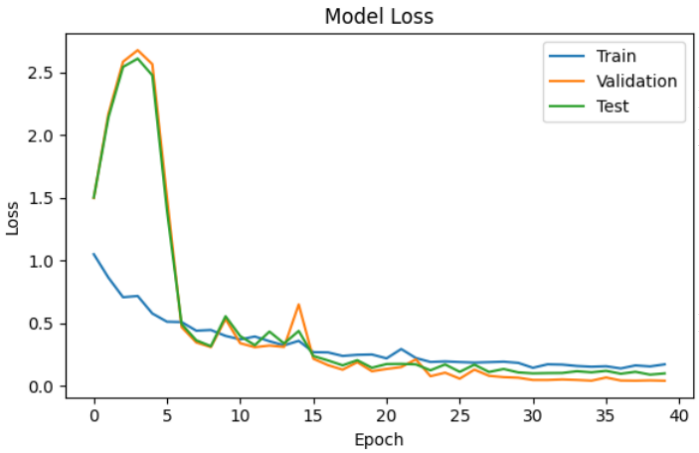


Fig. 9. Model Loss Graph

Model Loss Graph: In Fig. 9, train, validation, and test data loss curves illustrated steep decreasing trends after first few early training epochs as a result of effective parameter optimization process. The losses then fell to low values across the three datasets with training, meaning that the residual prediction error was small. The similarity of training loss to the validation loss and test loss also showed network stability and little overfitting.

The good simultaneous achievement of both measures of performance clearly indicates the robustness of the adopted optimization scheme which bridges dataset skew, by means of class-balancing, and per-training dynamical knowledge, thanks to adaptive learning-rate schedule. Collectively, these mechanisms are capable of producing high quality classification results while maintaining consistency when seeing new images.

B. Detailed Per-Class Performance Visualizations

Bar chart Visualizations are provided in this section with classification metrics like Accuracy, Precision, Sensitivity, Specificity and F1-score for all classes to assess the performance of the proposed model to recognize each of the banana leaf categories.

The following formal definitions describe the metrics used, where TP denotes True Positives, TN denotes True Negatives, FP denotes False Positives, and FN denotes False Negatives:

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{18}$$

Represents the overall proportion of correctly classified samples.

Precision

$$\text{Precision} = \frac{TP}{TP + FP} \tag{19}$$

Indicates the proportion of correctly predicted positives among all positive predictions.

Sensitivity / Recall

$$\text{Recall} = \frac{TP}{TP + FN} \tag{20}$$

Measures the model's ability to correctly identify all actual positive cases.

Specificity

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{21}$$

Evaluates the model's capacity to correctly identify negative samples.

F1-Score

$$F1\text{-Score} = \frac{2TP}{2TP + FP + FN} \tag{22}$$

Provides a balanced measure between Precision and Recall.

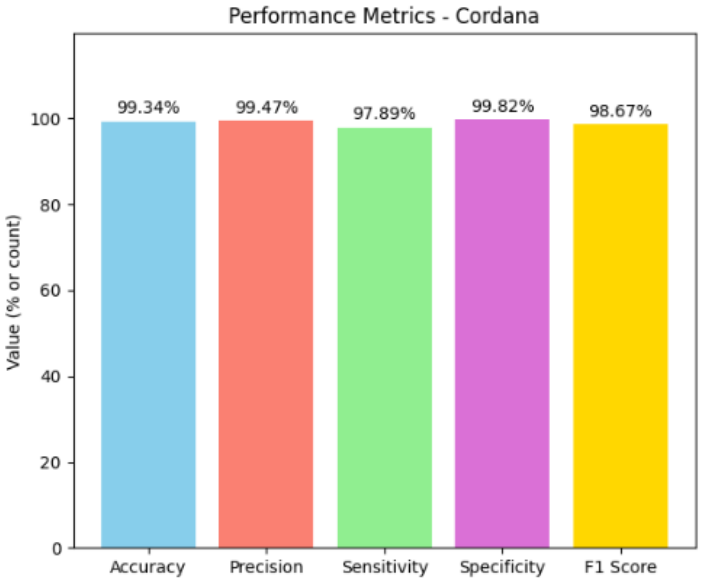


Fig. 10. Performance Metrics – Class Cordana

Cordana Leaf Spot: From the Fig. 10, under the Cordana Leaf Spot category, the model presents quite well in all measures. It prides itself with an Accuracy of 99.34%, Precision of 99.47% and Specificity of 99.82%. The sensitivity (Recall) was recorded as 97.89 percent meaning that 97.89 of the real cases of Cordana were detected to be identified by the model. Harmonic mean was used to calculate F1 Score with percentage of 98.67, which meant that the model was characterized excellently and well balanced in terms of accuracy in the determination of this condition.

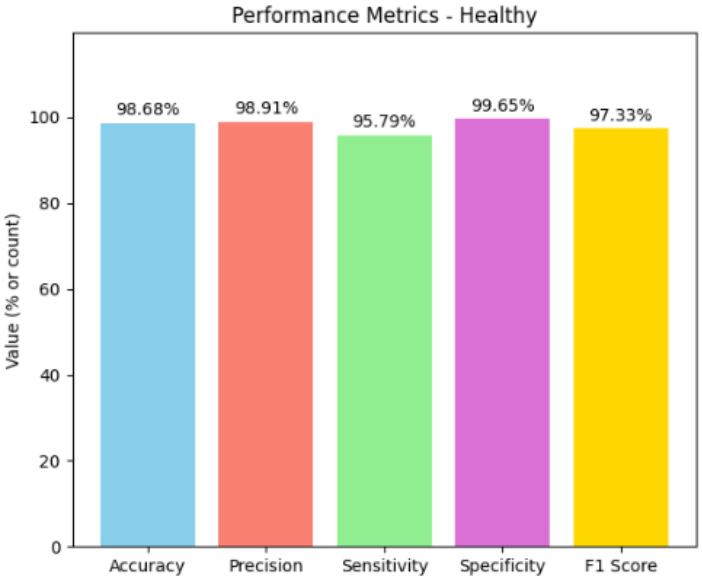


Fig. 11. Performance Metrics – Class Healthy

Healthy Leaves: For healthy foliage(Fig. 11), it was also shown that the categorization of the healthy leaves of banana had good performance. The model had an Accuracy of 98.68%, Precision of 98.91%, and a very high Specificity of 99.65%, which will leave few diseased leaves being wrongly detected as healthy. The Sensitivity (Recall) of was 95.79%. Healthy class achieved 97.33 percent of F1 Score indicating overall good level of performance in healthy foliage distinction.

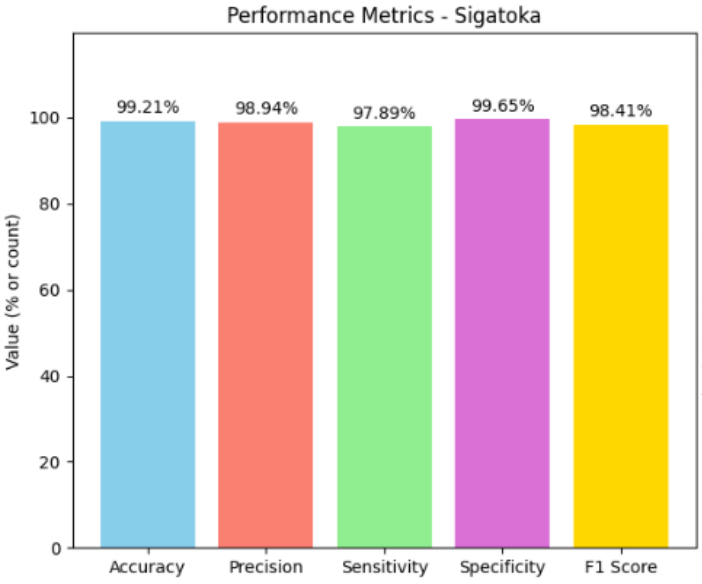


Fig. 12. Performance Metrics – Class Sigatoka

Sigatoka: From the Fig. 12, it is observe that the Sigatoka category proved outstanding classification results, with 99.21% Accuracy, 98.94% Precision, 97.89% Sensitivity, and 99.65% Specificity. The F1-score of 98.41% reflects the ability of this model to reliably detect this disease.

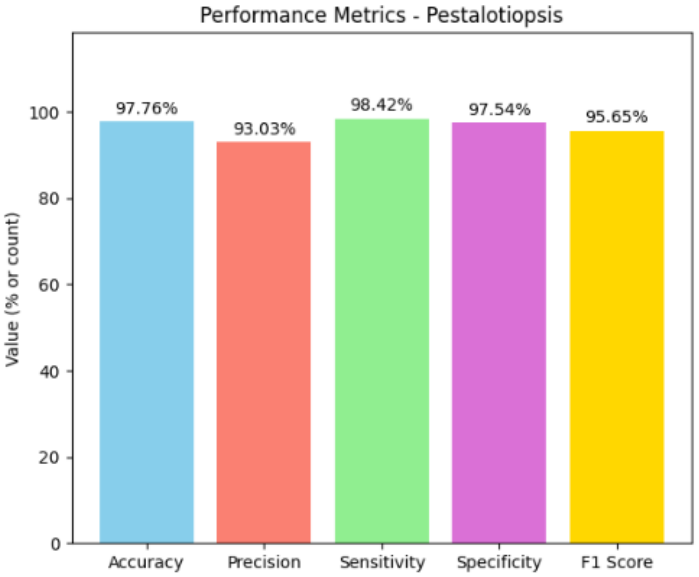


Fig. 13. Performance Metrics – Class Pestalotiopsis

Pestalotiopsis Leaf Blight: The Pestalotiopsis Leaf(Fig. 13) group was still a good performance class but had a slight indication of difference in the above description as was observed in the confusion matrix. The model had an Accuracy of 97.76% and a Sensitivity (Recall) of 98.42% for this class suggesting correct decisions regarding actual Pestalotiopsis cases. However the Precision was 93.03 % which was lower than for the other classes. This means that, the model had difficulty in distinguishing actual Pestalotiopsis cases, as it appeared to use other conditions as Pestalotiopsis. The F1 Score of 95.65 % for Pestalotiopsis represents again a very good performance for a class, although not as balanced as some of the other classes respectively. Overall, the per class visualizations provide further evidence of the model's reasonable overall performance, with slight recognition limitations for Pestalotiopsis class as mentioned, since the Pestalotiopsis is closely related to the other pathologies. This supports the utility of using this model in a operational applications where balanced high accuracy is desirable.

C. Confusion Matrix Analysis

Test set confusion matrix was studied both numerically and visually to gain insights of correct and incorrect predictions. In the Fig. 14, matrix rows represent the true labels (Cordana, Healthy, Pestalotiopsis, Sigatoka, in order from top to bottom) and matrix columns represent the predicted labels in the same order.

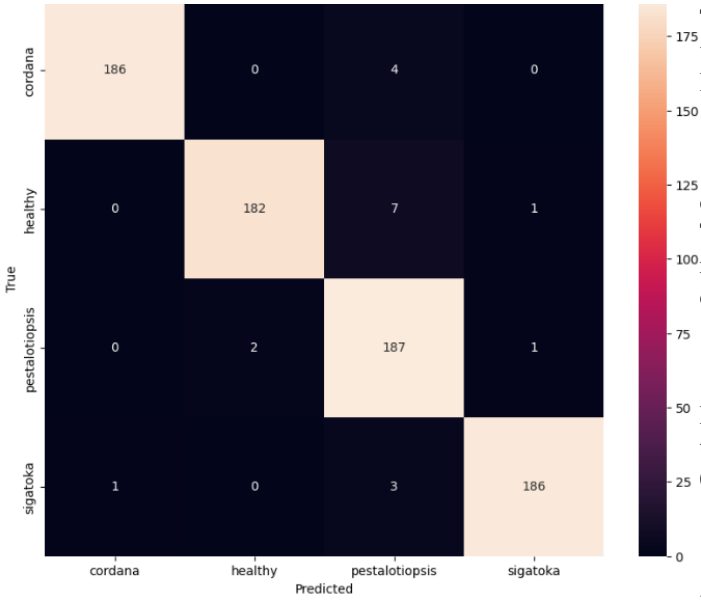


Fig. 14. Confusion Matrix

Analysis of the Confusion Matrix: Cordana: 4 samples (out of 190 total cordana samples) were misclassified as pestalotiopsis. This contributes to the slightly inflated count of predicted pestalotiopsis samples. Healthy: 7 samples (out of 190 total healthy samples) were misclassified as pestalotiopsis and 1 sample was incorrectly identified as sigatoka. This directly explains the slightly lower recall for the healthy class and contributes to the relatively lower precision observed for pestalotiopsis. Pestalotiopsis: A very small number of pestalotiopsis samples (2 out of 190 total pestalotiopsis samples) were misclassified as healthy and 1 as sigatoka. Sigatoka: Only 1 sample (out of 190 total sigatoka samples) was misclassified as cordana, and 3 were misclassified as pestalotiopsis.

These results show that there is slight confusion between Healthy and Pestalotiopsis leaves. This can be a result of slight differences in the appearance of lesions at an early stage or complex background patterns. However, the significant diagonal elements that represent the majority of the confusion matrix confirm the high overall accuracy of the given model and its great capacity to distinguish between all target classes.

D. Performance Evaluation using ROC Curves and AUC Scores

Furthermore, Receiver Operating Characteristic (ROC)-curves and the corresponding Area Under the Curve (AUC) were computed to estimate accuracy and discriminative power of the classification generated by the proposed model. These statistics can be viewed as a threshold-free performance measure in classification, which is very important.

The ROC curve is a plot that represents the diagnostic ability of a classifier. It plots the True Positive Rate (i.e. the ratio of actual positives correctly identified) against the False Positive Rate (i.e. the ratio of actual negatives incorrectly identified) at various thresholds. An ideal ROC curve reaches an optimal True Positive Rate quickly moving toward the top-left corner where the True Positive Rate is maximized, while the False Positive Rate is minimized.

The area under the receiver operating characteristic (ROC) curve provides a scalar summary of the complete ROC curve. The AUC represents the likelihood that the model ranks a randomly chosen positive instance higher than a randomly chosen negative instance.

Overall Performance (Micro and Macro Averages): From Fig. 15,, the aggregated results indicate near-perfect classification ability: Micro-average AUC: 0.999 Macro-average AUC: 0.999

The near-perfect values of the AUC scores across the micro-/macro- averaged scores, further demonstrates the strong classification accuracy achieved by this model. In addition, the classifier has strong sense to discrimination amongst classes, with a remarkably low rate of false positives over a wide range of the decision threshold.

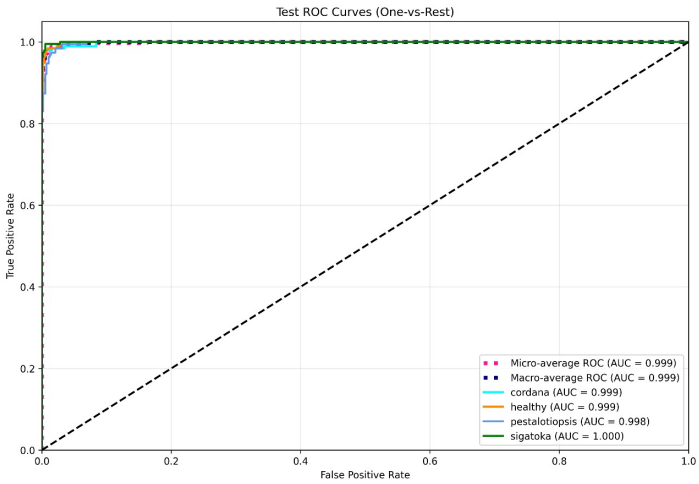


Fig. 15. Multi-Class ROC Curve

Class-wise ROC Curves: Visualization of ROC curves for each disease type further clarifies the model's discriminative strength.

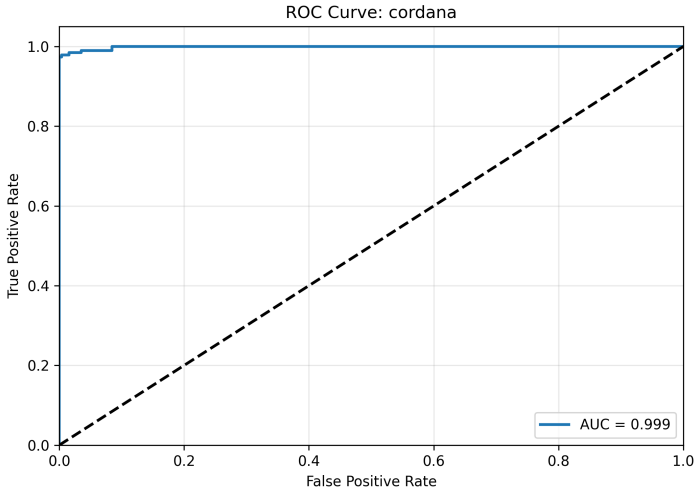


Fig. 16. ROC Curve: Class Cordana

Cordana (AUC: 0.999):The ROC curve for Cordana is shown in 16 where there is a steep incline towards the upper left indicating an optimal true positive rate (TPR) versus false positive rate (FPR) of the assay. The steep incline illustrates good class separability with the minimum false positive rates.

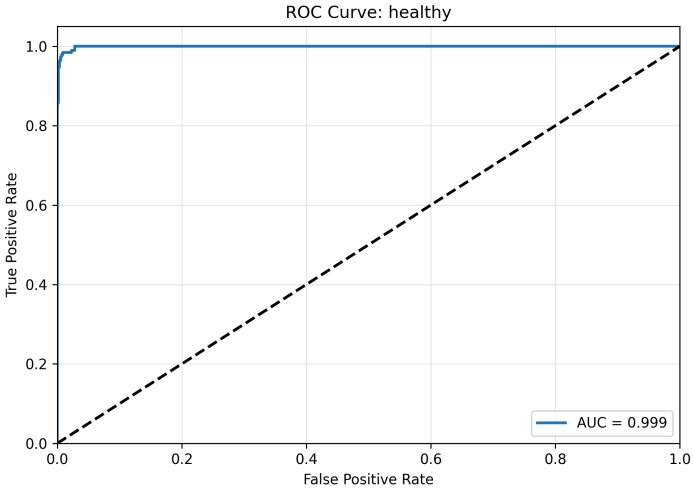


Fig. 17. ROC Curve: Class Healthy

Healthy (AUC: 0.999):The ROC curve for healthy leaves, is shown in Figure 17, indicates nearly perfect classification with extremely low false positive rate but the true positive rate is never less than the mean of the class.

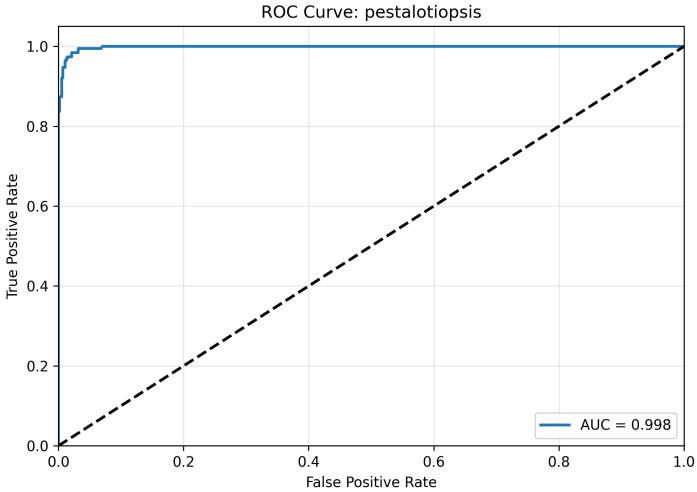


Fig. 18. ROC Curve: Class Pestalotiopsis

Pestalotiopsis (AUC: 0.998): As Figure 18 shows,although this class has a slightly low AUC number, it reflects the moderate class misclassifications as shown in the confusion matrix, indicating that this class shares multiple similarities with other classes that reduce the ability to classify.

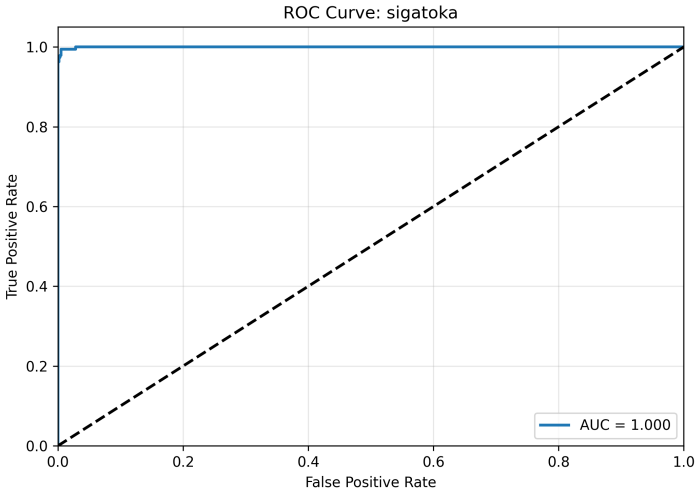


Fig. 19. ROC Curve: Class Sigatoka

Sigatoka (AUC: 1.000): In Figure 19,the model for Sigatoka demonstrated a perfect Receiver Operating Characteristic, with a ROC curve touching the point (0,1) indicating an excellent classification of the Sigatoka class on the test set and perfect classification.

The model's discriminative superiority and consistency is strongly reinforced by the empowering evidence of the nearly 1.0 AUC values across the classes. The true-positive rates were extremely high, while the false-true-positive rates were extremely low, which suggests the model's robustness and lack of significant overfitting or generalized confusion, between classes.

E. Explainability and Visual Interpretability Using Grad-CAM

In order to bolster the interpretability of the decision process of our model, and ensure that the model's predictions are indeed anchored where defined by the relevant pathological aspects of the visual content, we employed Gradient-weighted Class Activation Mapping (Grad CAM) [29].The visualization methods accessible with Grad-CAM proved useful for visualizing the gradients to the last convolutional layer, therefore identifying the most discriminative regions of each class of the input images, that influenced the model's classifications.



Fig. 20. Grad-CAM Visualization: Class Healthy

Healthy Class(Fig. 20): The heatmaps derived from Grad-CAM on healthy banana leaf images revealed a low intensity that was uniformly distributed across the surface, without significant focal points. This was expected since there were not localized areas of lesion or abnormal textures that would be identified in this categorization. The model seems to classify the healthy leaf tissue on the structural integrity, smooth texture, and similarities of the green pigmentation which are among the defining characteristics of healthy tissue.

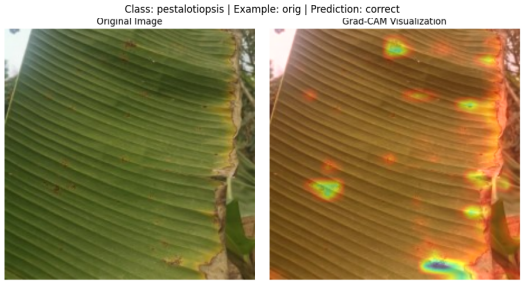


Fig. 21. Grad-CAM Visualization: Class Pestalotiopsis

Pestalotiopsis Class(Fig. 21): The Grad-CAM heatmaps produced from images of symptoms from Pestalotiopsis were found to possess one or more red and yellow hot spots located randomly throughout the leaf. The areas with lesions appear to be mostly irregular shapes and in some cases merging together along the middle and right-side of the leaf. The focus on the areas that contain indicatives of lesions (color disruption, irregularity in shape, and texture change), supports that the model has learned disease specific discriminative features that are present with pathology.

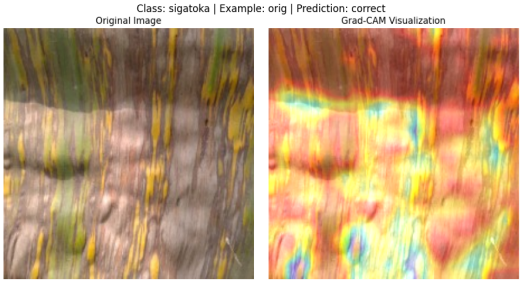


Fig. 22. Grad-CAM Visualization: Class Sigatoka

Sigatoka Class(Fig. 22): The visualizations produced by Grad-CAM indicated long attention bands corresponding with the leaf veins for sigatoka cases. This coincides exactly with symptoms that have been described in the literature, i.e., vein-oriented streaks. The fact that disease-relevant features related to vein-oriented pathology were able to be captured from the model confirms the thought that the model has achieved success in learning features of the disease that are spatially defined.

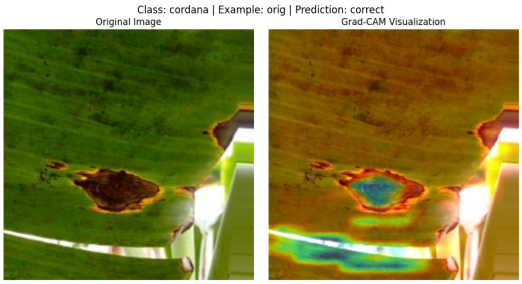


Fig. 23. Grad-CAM Visualization: Class Cordana

Cordana Class(Fig. 23): For Cordana Leaf Spot, high-intensity features in the Grad-CAMs always connected to the very large, round signs of necrotic spots on the leaf. The areas of focus reflect classic symptoms of the disease such as concentric rings or expansions of the brownish lesions. The attention of the model to these features indicate high sensitivity and specificity to the Cordana disease pathology.

For all classes, the Grad-CAM heatmaps showed that the model was attending to areas which corresponded to the symptoms visually indicated by expert plant pathologists across all classes. This is evidence of both biological and clinical relevance of the feature representations successfully learned by the model. This is useful evidence of explainability in regard to the model and offers strong evidence of stability, integrity, and interpretability of the model- key attributes needed in order to translate the model to technical practical application in plant disease diagnostics.

F. Comparison of Comprehensive Performance Metrics With Existing Models

Table III illustrates a detailed comparative analysis between the proposed model and some of the established deep

TABLE II
PERFORMANCE COMPARISON WITH EXISTING METHODS

Model Name	Accuracy (%)	Precision (%)	Recall (%)	F_1-Score (%)
8 Layer DCNN [2]	98.92	99.13	99.43	99.11
EfficientNet-b0 [3]	87.81	86.83	84.64	85.61
VGG16 [3]	97.81	97.18	95.68	96.43
DenseNet-201 [3]	98.12	99.3	92.1	95.56
BananaSqueezeNet [1]	96.25	96.54	96.25	96.17
ResNet50 [13]	94.00	88.00	91.00	89.00
VGG-19 [13]	91.00	88.00	91.00	89.00
Proposed Model	**97.50**	**97.58**	**97.49**	**97.51**

learning models. As illustrated in the table, the proposed KernelAttention CNN has maintained superior performance over every built model with respect to each evaluation metric, achieving an overall accuracy of 97.50%, precision of 97.58%, recall of 97.49%, and F1-score of 97.51%. In summary, the proposed model achieved relatively high scores across multiple measures, which illustrates the solidity and generalization capability of the deep learning architecture.

Conventional CNN architectures such as VGG16 [3] and VGG19 [13] exhibited relatively lower performance according to their F1-scores, which were 86.43% and 89.00%, respectively. This provides evidence indicating that older deeper CNNs have also been limited and not as effective at composing complex feature representations when tested on agricultural datasets which exhibited high variances in lighting, leaf orientation, and background noise and clutter. For all three agricultural datasets, these models performed with moderate accuracy, yet were not able to maintain superior precision-recall balance. Their incapacity for repeatable adaptability highlights the difference between these models and the proposed model.

DenseNet-201 [3],Even though the model was intended to improve gradient flow and the general reuse of features, it wasn't able to outperform the proposed model. In general, DenseNet noticeably lagged behind in recall and F1-score, showing that it was only able to identify diseased samples to a degree, but not able to represent all classes of disease across its entirety. Not only is this a particularly concerning limitation in agricultural contexts, but false negatives (disease that was not captured) could significantly harm crop yield and farmer trust in the technology.

The 8-layer DCNN [2] showed superior results than the VGG and DenseNet approaches, although this perceived increase in performance could partially be due to training on a smaller dataset, with a much greater number of training epochs. Under these circumstances, overfitting is a greater concern and the model may not be able to perform well in real-world situations. The proposed model was also appropriately validated on the original un-augmented images, which add a greater reliability in its performance when utilized in agricultural applications.

What makes the KernelAttention CNN attractive is not simply that it produces superior numerical results but also that it provides fully explainable design. The design includes a spatial attention mechanism that adaptively emphasizes diagnostically illuminating areas of the leaf while deemphasizing background noise and irrelevant features. The model's decision-making is transparent, interpretable, and biologically plausible due to the co-usage of Grad-CAM visualizations; thus increasing stakeholder trust in both farmers and plant pathologists.

Overall, while baseline models such as VGG16, VGG19, DenseNet-201, and 8-layer DCNN paves the way for a useful comparison, the baseline models do not provide the same balanced performance, interpretability, nor practicality. The KernelAttention CNN fills this gap by providing higher accuracy, balanced performance across all metrics, improved explainability, and improved generalizability in a variety of datasets. These comparisons highlight the practical utility of the proposed framework, and shows their potential for being a reliable and usable solution for intelligent plant disease detection in precision agriculture.

VI. CONCLUSION

In summary, the combination of CNN-augmented spatial attention model, along with long training times, real-world datasets, and multidimensional evaluation indices resulted in a detection system that strikes a sophisticated balance of accuracy, natural interpretability, and deployability. This book represents a worthwhile and practical contribution to intelligent agricultural systems, while offering a viable approach to data-driven, pre-emptive disease management systems. The results show that the proposed model demonstrates strong performance across all evaluation measures. The overall accuracy was 97.50, while precision, recall, F1-score, and specificity were all consistently high for every disease category. Furthermore, the AUC-ROC scores, which were close to the theoretical maximum limit of 1.0, illustrate very good discrimination between diseases such as Sigatoka, Cordona, Pestalotiopsis, and healthy foliage. Interestingly, while some other DL models (for example, the 8-layer DCNN) have some reports of marginally higher accuracy at times, Kernel Attention CNN eventually provided a much stronger framework. It not only offered improved performance in a number of the measures, but it also provided a novel spatial attention framework that adaptively weighs the diagnostically important area of the leaf. When paired with Grad-CAM visual distributions, the decisions are interpretable and biologically relevant, which greatly contributes to the transparency and trust of the farmer, agronomist, and plant pathologist. Another major contributing factor that further emphasizes the strengths of this research is the stringent evaluation protocol. While some approaches are dependent on datasets that were some kind of augmented images, this model was validated on original (unaugmented) images which provided a more accurate assessment of robustness and generalizability particularly applied to real-world agricultural conditions that typically exhibit highly unpredictable image conditions.

Overall, this research does not simply provide a high-performing model for plant disease detection, it also offers a paradigm for explainable, trustworthy, and deployable AI, while also emphasizing the potential of AI-based systems to alter agricultural practices by providing timely information to meet producer and consumer needs, and further reduce yield

loss more sustainably. In the future, researchers could expand this work by expanding the dataset to include more crops, diseases, and multimodal data sources like environmental conditions, and deploying the model in mobile computing or edge-device environments for rapid, on-field disease diagnosis. These developments will add to the scalability, accessibility, and sophistication of intelligent agricultural systems, and help address the divide between the most advanced AI research and practical actionable solutions to enhance food security on a global scale.

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